

12.6 Appendix

Appendix 12.A: Correlation Coefficient of Two Signals with Time and Frequency Separation

In the following, we compute the correlation coefficient of two signals that have a temporal separation τ and a frequency separation $f_1 - f_2$. The temporal distance can be easily converted into a spatial separation, so that the equation is quite general: it can be applied to spatial, temporal, and frequency diversity. The derivation closely follows [Jakes 1974].

We first define four real Gaussian variables $x_1 \dots x_4$

$$\begin{aligned} x_1 &= E_0 \sum_{\text{angle } n} \sum_{\text{delay } l} a_{nl} \cos [2\pi (\nu_n t - f_1 \tau_{nl})] \\ x_2 &= E_0 \sum_{\text{angle } n} \sum_{\text{delay } l} a_{nl} \sin [2\pi (\nu_n t - f_1 \tau_{nl})] \\ x_3 &= E_0 \sum_{\text{angle } n} \sum_{\text{delay } l} a_{nl} \cos [2\pi (\nu_n (t + \tau) - f_2 \tau_{nl})] \\ x_4 &= E_0 \sum_{\text{angle } n} \sum_{\text{delay } l} a_{nl} \sin [2\pi (\nu_n (t + \tau) - f_2 \tau_{nl})] \end{aligned} \quad (12.55)$$

where ν_n is the Doppler-shift of the n -th partial wave (with angle-of-incidence α_n), and this wave consists of L contributions with delays τ_{nl} . The a_{nl} are the amplitudes of the partial waves, and f_1 and f_2 are the two considered (angular) frequencies. The joint pdf of four zero-mean Gaussian variables is

$$pdf_{x_1, x_2, x_3, x_4} (x_1, x_2, x_3, x_4) = \frac{1}{(2\pi)^2 \Delta_R^{1/2}} \exp \left(-\frac{1}{2} \mathbf{x}^T \mathbf{R}^{-1} \mathbf{x} \right) \quad (12.56)$$

where $\mathbf{R}$ is the correlation matrix, and Δ_R its determinant. Transformation to polar coordinates gives

$$pdf_{r_1, r_2, \psi_1, \psi_2} (r_1, r_2, \psi_1, \psi_2) = \frac{r_1 r_2}{(2\pi R_{11})^2 (1 - \kappa^2)} \cdot \exp \left[-\frac{r_1^2 + r_2^2 - 2r_1 r_2 \kappa \cos (\psi_2 - \psi_1 - \varsigma)}{2R_{11} (1 - \kappa^2)} \right] \quad (12.57)$$

where by definition

$$\tan \varsigma = \frac{R_{14}}{R_{13}} = \frac{E \{x_1 x_4\}}{E \{x_1 x_3\}} \quad \kappa^2 = \frac{R_{13}^2 + R_{14}^2}{R_{11}^2} = \frac{E \{x_1 x_3\}^2 + E \{x_1 x_4\}^2}{(E \{x_1^2\})^2} \quad R_{11} = E \{x_1^2\}. \quad (12.58)$$

The statistical properties are thus determined by the correlation coefficients $R_{13} = E \{x_1 x_3\}$, $R_{14} = E \{x_1 x_4\}$, $R_{11} = E \{x_1^2\}$. Interchanging of summation and expectation gives

$$R_{11} = E_0^2 \sum_{n, l, p, q} E \{a_{nl} a_{pq} \cos [2\pi (\nu_n t - f_1 \tau_{nl})] \cos [2\pi (\nu_p t - f_1 \tau_{pq})]\} \quad (12.59)$$

The expectation of the products of the cosines is $1/2$, if $n = p$ and $l = q$ and otherwise 0

$$R_{11} = \frac{E_0^2}{2} \sum_{n, l} a_{nl}^2 = \Omega_0 \cdot \sum_{n, l} G_{RX}(\alpha_n) pdf(\alpha_n, \tau_{nl}) d\alpha dt \quad (12.60)$$

where

$$\Omega_0 = \int S(f) df \quad (12.61)$$

and $S(f)$ is the Doppler spectrum of the received signal. $G_{RX}(\alpha)$ is the antenna power pattern of the receive antenna, and $pdf_{\alpha,t}$ is the spatio-temporal distribution of the received signal. Since there are many partial waves, the summation in Eq. (12.60) can be turned into an integration

$$R_{11} = \Omega_0 \int_0^{2\pi} G_{RX}(\alpha) d\alpha \int_0^\infty pdf_{\alpha,t}(\alpha, t) dt. \quad (12.62)$$

Analogously, we get

$$\begin{aligned} R_{13} &= \Omega_0 \int_0^{2\pi} G_{RX}(\alpha) \int_0^\infty pdf_{\alpha,t}(\alpha, t) \cos(k_0 v \tau \cos \alpha - 2\pi(f_2 - f_1)t) dt d\alpha \\ R_{14} &= \Omega_0 \int_0^{2\pi} G_{RX}(\alpha) \int_0^\infty pdf_{\alpha,t}(\alpha, t) \sin(k_0 v \tau \cos \alpha - 2\pi(f_2 - f_1)t) dt d\alpha, \end{aligned} \quad (12.63)$$

where k_0 is the wavenumber, and we used the fact that the Doppler shift is proportional to the wavenumber k_0 , the velocity v and to the directional cosine: $\nu = k_0 v \cos \alpha$.

If the power delay profile is an exponential function, $\exp(-t/S_\tau)$, we use an omni-antenna, and the signal is isotropically incident from all directions, then

$$R_{11} = \Omega_0 \quad (12.64)$$

$$\begin{aligned} R_{13} &= \Omega_0 \int_0^{2\pi} \frac{1}{2\pi} \cos(k_0 v \tau \cos(\alpha)) \int_0^\infty \frac{1}{S_\tau} \exp(-t/S_\tau) \cos[2\pi(f_2 - f_1)t] dt d\alpha \\ &\quad + \Omega_0 \int_0^{2\pi} \frac{1}{2\pi} \sin(k_0 v \tau \cos(\alpha)) \int_0^\infty \frac{1}{S_\tau} \exp(-t/S_\tau) \sin[2\pi(f_2 - f_1)t] dt d\alpha \\ &= \Omega_0 \frac{J_0(k_0 v \tau)}{1 + (2\pi)^2 S_\tau^2 (f_2 - f_1)^2}. \end{aligned} \quad (12.65)$$

$$\begin{aligned} R_{14} &= \Omega_0 \int_0^{2\pi} \frac{1}{2\pi} \sin(k_0 v \tau \cos(\alpha)) \int_0^\infty \frac{1}{S_\tau} \exp(-t/S_\tau) \cos[2\pi(f_2 - f_1)t] dt d\alpha \\ &\quad - \Omega_0 \int_0^{2\pi} \frac{1}{2\pi} \cos(k_0 v \tau \cos(\alpha)) \int_0^\infty \frac{1}{S_\tau} \exp(-t/S_\tau) \sin[2\pi(f_2 - f_1)t] dt d\alpha \\ &= -2\pi(f_2 - f_1) S_\tau \Omega_0 \frac{J_0(k_0 v \tau)}{1 + (2\pi)^2 S_\tau^2 (f_2 - f_1)^2} \end{aligned} \quad (12.66)$$

and therefore

$$\kappa^2 = \frac{J_0^2(k_0 v \tau)}{1 + (2\pi)^2 S_\tau^2 (\omega_2 - \omega_1)^2} \quad (12.67)$$

For the computation of the envelope correlation coefficient, we first need

$$R_{12} = E\{r_1 r_2\} = \int_0^\infty \int_0^\infty r_1 r_2 \int_0^{2\pi} \int_0^{2\pi} pdf_{r_1, r_2, \psi_1, \psi_2}(r_1, r_2, \psi_1, \psi_2) d\psi_1 d\psi_2 dr_1 dr_2. \quad (12.68)$$

Integration over ψ_1 and ψ_2 gives

$$\begin{aligned} \int_0^\infty \int_0^\infty r_1 r_2 \cdot \frac{r_1 r_2}{R_{11}^2 (1 - \kappa^2)} \cdot \exp\left[-\frac{r_1^2 + r_2^2}{2R_{11} (1 - \kappa^2)}\right] I_0\left(\frac{r_1 r_2}{R_{11}} \frac{\kappa}{1 - \kappa^2}\right) dr_1 dr_2 \\ = \frac{\pi}{2} \Omega_0 F_{\text{hyper}}\left(-\frac{1}{2}, -\frac{1}{2}, 1, \kappa^2\right) \end{aligned} \quad (12.69)$$

where F_{hyper} is the hypergeometric function. This exact result can be approximated as

$$E \{r_1 r_2\} \approx \frac{\pi}{2} \Omega_0 \left(1 + \frac{\kappa^2}{4} \right) . \quad (12.70)$$

The computation of the covariance

$$\frac{E \{r_1 r_2\} - E \{r_1\} \cdot E \{r_2\}}{\sqrt{\left[E \{r_1^2\} - E \{r_1\}^2 \right] \cdot \left[E \{r_2^2\} - E \{r_2\}^2 \right]}} \quad (12.71)$$

furthermore uses

$$\begin{aligned} E \{r_1^2\} &= 2\Omega_0 \\ E \{r_1\} &= \sqrt{\frac{\pi\Omega_0}{2}}, \end{aligned} \quad (12.72)$$

which follows from the fact that r_1, r_2 are Rayleigh-distributed. Thus,

$$\rho = \kappa^2. \quad (12.73)$$

Inserting for κ^2 we obtain the final result

$$\rho_{xy} = \frac{J_0^2(k_0 v \tau)}{1 + (2\pi)^2 S_r^2 (f_2 - f_1)^2}. \quad (12.74)$$

This is the equation that most textbooks give for the covariance, and that we use extensively in Sec. 12.2. Note, however, that many restrictive assumptions were required in order to arrive at this simple equation. It is assumed a priori that the WSSUS model is valid, no LOS exists, the power delay profile has an exponential shape, the incident power is isotropically distributed in azimuth and only propagates in the horizontal plane, and an omni-antenna is used.